

Sweedel Rodrigues

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SUMMARY

AI/ML Engineer with hands-on experience in computer vision, 3D reconstruction, and full-stack applications. Experienced in designing scalable AI systems, with exposure to YOLO-based detection, retrieval-augmented generation (RAG), and anomaly detection. Driven to build impactful, real-world intelligent solutions.

EDUCATION

Dayananda Sagar College of Engineering
B.E. Information Science

Sept 2027
Current GPA: 9.6 / 10.0

SKILLS

Programming Languages: Python, C, C++, JavaScript
Web & Backend Development: MERN Stack, REST APIs
Databases: MySQL, MongoDB, PostgreSQL, PHP
AI / Machine Learning: NumPy, Pandas, Scikit-learn, YOLO
Tools & Platforms: Git, GitHub, Postman, VS Code, Linux

EXPERIENCE

GREP 3D Reconstruction System

Sept 2025 – March 2026

3D Computer Vision Intern

- Designed and built an end-to-end 3D reconstruction pipeline from multi-view images by performing feature extraction, feature matching, and camera pose estimation, enabling the generation of accurate and structured point clouds for realistic spatial representation of complex real-world scenes.
- Leveraged COLMAP and Fast3R-based reconstruction techniques to perform robust image alignment, sparse-to-dense mapping, and geometric consistency optimization, improving reconstruction accuracy, reducing noise, and enhancing the overall quality and reliability of generated 3D models.
- Applied geometric segmentation and SAM-based methods to refine reconstructed scenes by isolating meaningful structures, enabling precise structural alignment, noise filtering, and improved spatial analysis for better interpretation of reconstructed environments.

PROJECTS

MediScan | React, Node.js, Express, PostgreSQL | *GitHub*

- Built a full-stack medicine authentication system using QR-based verification to detect counterfeit drugs and improve transparency across the pharmaceutical supply chain.
- Enabled quick and reliable access to verified medicine information through QR scanning, improving user trust and ensuring safer pharmaceutical usage.
- Designed and implemented a secure backend architecture for managing medicine data, ensuring data integrity and efficient handling of authentication requests.

CarbonEdge | *Python, Flask, LSTM* | *GitHub*

- Built an AI-driven industrial decarbonization platform to detect hidden energy inefficiencies using multi-sensor data streams across industrial systems.
- Designed anomaly detection and energy behavior modeling pipelines to identify subtle drifts and generate actionable insights for reducing energy consumption and operational costs.

ACHIEVEMENTS

Published a research paper titled “*MediScan – Know Before You Dose: Scan QR Codes to Track Expiry Dates and Unlock Trusted Medicine Details*” at the **9th IEEE International Conference on Computational Systems & Information Technology for Sustainable Solutions (CSITSS 2025)**.

Secured **1st Place** at **HackBites 3.0**, a 24-hour hackathon organized by **Global Academy of Technology, Bengaluru**.